

Conjecture on the Number of Linearly Independent Relations of Ihara–Kaneko–Zagier Derivation Relation: An Euler-type Pentagonal Formula

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1. GENERALIZED PENTAGONAL NUMBERS AND THE $\omega(n)$ FUNCTION

Let $P_0, P_1, P_2, P_3, P_4, P_5, P_6, P_7, \dots = 0, 1, 2, 5, 7, 12, 15, 22, \dots$ denote the generalized pentagonal numbers OEIS:A001318, and let $\omega(k) = A010815(k)$, defined by

$$\omega(k) = \begin{cases} 1 & \text{if } k = 0; \\ (-1)^m & \text{if } k = \frac{3m^2 \pm m}{2}; \\ 0 & \text{otherwise.} \end{cases}$$

Table 1. P_n and $\omega(n)$

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
P_n	0	1	2	5	7	12	15	22	26	35	40	51	57	70	77	92	100	117	126	145	155	176	187	210
$\omega(n)$	1	-1	-1	0	0	1	0	1	0	0	0	0	-1	0	0	-1	0	0	0	0	0	0	1	0

2. THE CONJECTURE AND SOME EQUIVALENT FORMULATIONS

For a given weight $n \geq 3$ let $\mathcal{L}(n)$ denote the the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation. In [1, 2] the authors have tabulated the values of $\mathcal{L}(n)$ calculated using the Risa/Asir and Mathematica general computer algebra system for $n = 3$ to $n = 15$.

Table 2. Number of Independent Derivation Relations for MZVs

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16*	17*
$\mathcal{L}(n)$	1	2	5	10	22	44	90	181	363	727	1456	2912	5825	11652*	23304*

* Calculated with a JavaScript application of my own development.¹

Conjecture 1. *For all weight $n \geq 3$ the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation can be calculated by the following Euler-type pentagonal formula:*

$$\mathcal{L}(n) = 2^{n-2-P_1} + 2^{n-2-P_2} - 2^{n-2-P_3} - 2^{n-2-P_4} \pm \dots - \omega(n-1) - 1 = - \sum_{k=1}^{n-2} \omega(k) 2^{n-2-k} - \omega(n-1) - 1.$$

The generating function of $\mathcal{L}(n)$ is given by:

$$F(x) = \frac{x^2}{x-1} + \frac{x(x-1)}{2x-1} \prod_{k \geq 1} (1-x^k),$$

and $\mathcal{L}(n)$ satisfies the following recurrence relation for all $n \geq 3$:

$$\mathcal{L}(n+1) = 2 \cdot \mathcal{L}(n) + 1 - \omega(n) + \omega(n-1).$$

The calculations below verify that Conjecture 1. holds for $n = 3$ through $n = 17$:

$$\mathcal{L}(3) = 2^{1-1} - \omega(2) - 1 = 2^0 + 1 - 1 = 1 + 1 - 1 = 1$$

$$\mathcal{L}(4) = 2^{2-1} + 2^{2-2} - \omega(3) - 1 = 2^1 + 2^0 - 0 - 1 = 2 + 1 - 1 = 2$$

$$\mathcal{L}(5) = 2^{3-1} + 2^{3-2} - \omega(4) - 1 = 2^2 + 2^1 - 0 - 1 = 4 + 2 - 1 = 5$$

$$\mathcal{L}(6) = 2^{4-1} + 2^{4-2} - \omega(5) - 1 = 2^3 + 2^2 - 1 - 1 = 8 + 4 - 1 - 1 = 10$$

$$\mathcal{L}(7) = 2^{5-1} + 2^{5-2} - 2^{5-5} - \omega(6) - 1 = 2^4 + 2^3 - 2^0 - 0 - 1 = 16 + 8 - 1 - 1 = 22$$

¹ On page <https://tamas-tech.github.io>

$$\begin{aligned}
\mathcal{L}(8) &= 2^{6-1} + 2^{6-2} - 2^{6-5} - \omega(7) - 1 = 2^5 + 2^4 - 2^1 - 1 - 1 = 32 + 16 - 2 - 1 - 1 = \mathbf{44} \\
\mathcal{L}(9) &= 2^{7-1} + 2^{7-2} - 2^{7-5} - 2^{7-7} - \omega(8) - 1 = 2^6 + 2^5 - 2^2 - 2^0 - 0 - 1 = 64 + 32 - 4 - 1 - 1 = \mathbf{90} \\
\mathcal{L}(10) &= 2^{8-1} + 2^{8-2} - 2^{8-5} - 2^{8-7} - \omega(9) - 1 = 2^7 + 2^6 - 2^3 - 2^1 - 0 - 1 = 128 + 64 - 8 - 2 - 1 = \mathbf{181} \\
\mathcal{L}(11) &= 2^{9-1} + 2^{9-2} - 2^{9-5} - 2^{9-7} - \omega(10) - 1 = 2^8 + 2^7 - 2^4 - 2^2 - 0 - 1 = 256 + 128 - 16 - 4 - 1 = \mathbf{363} \\
\mathcal{L}(12) &= 2^{10-1} + 2^{10-2} - 2^{10-5} - 2^{10-7} - \omega(11) - 1 = 2^9 + 2^8 - 2^5 - 2^3 - 0 - 1 = 512 + 256 - 32 - 8 - 1 = \mathbf{727} \\
\mathcal{L}(13) &= 2^{11-1} + 2^{11-2} - 2^{11-5} - 2^{11-7} - \omega(12) - 1 = 2^{10} + 2^9 - 2^6 - 2^4 + 1 - 1 = 1024 + 512 - 64 - 16 = \mathbf{1456} \\
\mathcal{L}(14) &= 2^{12-1} + 2^{12-2} - 2^{12-5} - 2^{12-7} + 2^{12-12} - \omega(13) - 1 = 2^{11} + 2^{10} - 2^7 - 2^5 + 2^0 - 0 - 1 = 2048 + 1024 - 128 - 32 + 1 - 1 = \mathbf{2912} \\
\mathcal{L}(15) &= 2^{13-1} + 2^{13-2} - 2^{13-5} - 2^{13-7} + 2^{13-12} - \omega(14) - 1 = 2^{12} + 2^{11} - 2^8 - 2^6 + 2^1 - 0 - 1 = 4096 + 2048 - 256 - 64 + 2 - 1 = \mathbf{5825} \\
\mathcal{L}(16) &= 2^{14-1} + 2^{14-2} - 2^{14-5} - 2^{14-7} + 2^{14-12} - \omega(15) - 1 = 2^{13} + 2^{12} - 2^9 - 2^7 + 2^2 + 1 - 1 = 8192 + 4096 - 512 - 128 + 4 = \mathbf{11652} \\
\mathcal{L}(17) &= 2^{15-1} + 2^{15-2} - 2^{15-5} - 2^{15-7} + 2^{15-12} + 2^{15-15} - \omega(16) - 1 = 2^{14} + 2^{13} - 2^{10} - 2^8 + 2^3 + 2^0 - 0 - 1 = \\
&= 16384 + 8192 - 1024 - 256 + 8 + 1 - 1 = \mathbf{23304}
\end{aligned}$$

The first equivalent version reveals a simple connection between $\mathcal{L}(n)$ and $\text{pic}(n) = \text{A178841}(n)$ the number of pure inverting compositions of n .

Conjecture 2. For all weight $n \geq 3$ the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation can be calculated by the following formula:

$$\mathcal{L}(n) = 2^{n-2} - 1 - \text{pic}(n-1)$$

The calculations summarized in the table below verify that Conjecture 2. holds for $n = 3$ through $n = 17$:

Table 3. $\mathcal{L}(n) = 2^{n-2} - 1 - \text{pic}(n-1)$

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$2^{n-2} - 1$	1	3	7	15	31	63	127	255	511	1023	2047	4095	8191	16383	32767
$\text{pic}(n-1)$	0	1	2	5	9	19	37	74	148	296	591	1183	2366	4731	9463
$2^{n-2} - 1 - \text{pic}(n-1)$	1	2	5	10	22	44	90	181	363	727	1456	2912	5825	11652	23304

Let $c(n) = \text{A152537}(n)$. The second equivalent reformulation highlights a deeper algebraic relationship between $c(n)$ and $\text{pic}(n)$.

Conjecture 3. For all weight $n \geq 3$ the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation can be calculated by the following formula:

$$\mathcal{L}(n) = 2^{n-2} - 1 - \omega(n-1) - c(n-2)$$

The calculations summarized in the table below verify that Conjecture 3. holds for $n = 3$ through $n = 17$:

Table 4. $\mathcal{L}(n) = 2^{n-2} - 1 - \omega(n-1) - c(n-2)$

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$2^{n-2} - 1$	1	3	7	15	31	63	127	255	511	1023	2047	4095	8191	16383	32767
$\omega(n-1)$	-1	0	0	1	0	1	0	0	0	0	-1	0	0	-1	0
$c(n-2)$	1	1	2	4	9	18	37	74	148	296	592	1183	2366	4732	9463
$2^{n-2} - 1 - \omega(n-1) - c(n-2)$	1	2	5	10	22	44	90	181	363	727	1456	2912	5825	11652	23304

REFERENCES

- [1] Hideki Murahara. Derivation relations for finite multiple zeta values. *International Journal of Number Theory*, 13(02):419–427, 2017.
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